

Lecture 2 Planned Notes

Wednesday, May 21, 2025

3:18 PM

Review

Line integrals of the form:

$$(I) \quad \int_C f \, dt = \int_a^b f(\vec{r}(t)) \, dt \quad \text{where } C: \vec{r}(t), \quad a \leq t \leq b$$

$$(II) \quad \int_C f \, dl = \int_a^b f(\vec{r}(t)) |\vec{r}'(t)| \, dt \quad \text{where } C: \vec{r}(t), \quad a \leq t \leq b, \quad dl = |\vec{r}'(t)| dt$$

$$(III) \quad \int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) \, dt \quad \text{where } C: \vec{r}(t), \quad a \leq t \leq b, \quad d\vec{r} = \vec{r}'(t) dt$$

$$(IV) \quad \int_C \vec{F} \, dt = \int_a^b \vec{F}(\vec{r}(t)) \, dt, \quad \text{where } C: \vec{r}(t), \quad a \leq t \leq b, \quad \text{and the result is a vector}$$

THEOREM 1

Direction-Preserving Parametric Transformations

Any representations of C that give the same positive direction on C also yield the same value of the line integral (3).

PROOF

The proof follows by the chain rule. Let $\mathbf{r}(t)$ be the given representation with $a \leq t \leq b$ as in (3). Consider the transformation $t = \phi(t^*)$ which transforms the t interval to $a^* \leq t^* \leq b^*$ and has a positive derivative dt/dt^* . We write $\mathbf{r}(t) = \mathbf{r}(\phi(t^*)) = \mathbf{r}^*(t^*)$. Then $dt = (dt/dt^*) dt^*$ and

$$\begin{aligned} \int_C \mathbf{F}(\mathbf{r}^*) \cdot d\mathbf{r}^* &= \int_{a^*}^{b^*} \mathbf{F}(\mathbf{r}(\phi(t^*))) \cdot \frac{d\mathbf{r}}{dt} \frac{dt}{dt^*} dt^* \\ &= \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \frac{d\mathbf{r}}{dt} dt = \int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r}. \end{aligned}$$

THEOREM 2

Path Dependence

The line integral (3) generally depends not only on $\mathbf{F}$ and on the endpoints A and B of the path, but also on the path itself along which the integral is taken.

THEOREM 3

Path Independence

A line integral (1) with continuous F_1, F_2, F_3 in a domain D in space is path independent in D if and only if $\mathbf{F} = [F_1, F_2, F_3]$ is the gradient of some function f in D ,

$$(2) \quad \mathbf{F} = \text{grad } f, \quad \text{thus,} \quad F_1 = \frac{\partial f}{\partial x}, \quad F_2 = \frac{\partial f}{\partial y}, \quad F_3 = \frac{\partial f}{\partial z}.$$